

Wazuh Docker Single-Node Installation Guide (Ubuntu)

Environment

- Ubuntu 24.04
 - Docker Compose
 - Wazuh version: 4.8.2
-

1. Install Docker

Update packages:

```
sudo apt update
```

Install required packages:

```
sudo apt install -y ca-certificates curl gnupg lsb-release
```

Add Docker GPG key:

```
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -
o /etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

Add Docker repository:

```
echo
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/
docker.gpg]
  https://download.docker.com/linux/ubuntu
  $(. /etc/os-release && echo $VERSION_CODENAME) stable" |
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

Install Docker:

```
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin
docker-compose-plugin
```

Verify installation:

```
docker --version
docker compose version
```

2. Configure vm.max_map_count

Wazuh Indexer requires higher memory mapping limits.

Temporary change:

```
sudo sysctl -w vm.max_map_count=262144
```

Permanent change:

```
echo "vm.max_map_count=262144" | sudo tee -a /etc/sysctl.conf
```

Verify:

```
sysctl vm.max_map_count
```

Expected output:

```
vm.max_map_count = 262144
```

3. Download Wazuh Docker Repository

Create working directory:

```
mkdir -p ~/docker/wazuh  
cd ~/docker/wazuh
```

Clone repository:

```
git clone https://github.com/wazuh/wazuh-docker.git
```

Move into single-node directory:

```
cd wazuh-docker/single-node
```

4. Checkout Correct Wazuh Version

Switch to Wazuh 4.8.2:

```
git checkout v4.8.2
```

Explanation:

- The latest `main` branch may not contain required files.
- Version `v4.8.2` includes:
 - `generate-indexer-certs.yml`
 - correct configuration files
 - certificate generator

Verify files:

```
ls -l
```

Expected files:

```
config/  
docker-compose.yml  
generate-indexer-certs.yml  
README.md
```

5. Generate SSL Certificates

Generate all required certificates:

```
docker compose -f generate-indexer-certs.yml run --rm generator
```

Explanation:

This command creates:

- Root CA certificate
- Admin certificates
- Indexer certificates
- Dashboard certificates
- Filebeat certificates

Verify generated files:

```
ls config/wazuh_indexer/
```

Expected output:

```
internal_users.yml  
wazuh.indexer.yml
```

6. Start Wazuh Stack

Start all containers:

```
docker compose up -d
```

Explanation:

This starts:

- Wazuh Manager
- Wazuh Indexer
- Wazuh Dashboard

Verify containers:

```
docker compose ps
```

Expected output:

```
single-node-wazuh.dashboard-1  
single-node-wazuh.indexer-1  
single-node-wazuh.manager-1
```

7. Access Wazuh Dashboard

Open browser:

```
https://SERVER-IP
```

Example:

```
https://192.168.1.40
```

Important:

- Wazuh uses HTTPS.
- Ignore browser SSL warning.
- Click:

```
Advanced → Proceed
```

8. Wazuh Dashboard Login

Default credentials:

```
Username: admin  
Password: SecretPassword
```

Explanation of passwords in docker-compose.yml:

Variable	Purpose
INDEXER_PASSWORD	Main admin login password
DASHBOARD_PASSWORD	Internal dashboard service password
API_PASSWORD	Internal API communication password

Only use:

```
admin / SecretPassword
```

for browser login.

9. Useful Docker Commands

Check running containers

```
docker compose ps
```

View all logs

```
docker compose logs -f
```

View dashboard logs

```
docker compose logs -f wazuh.dashboard
```

View manager logs

```
docker compose logs -f wazuh.manager
```

View indexer logs

```
docker compose logs -f wazuh.indexer
```

Stop containers

```
docker compose stop
```

Start stopped containers

```
docker compose start
```

Restart containers

```
docker compose restart
```

Stop and remove containers

```
docker compose down
```

Remove everything including volumes

```
docker compose down -v
```

10. Common Issues and Fixes

Issue: vm.max_map_count too low

Error:

```
max virtual memory areas vm.max_map_count [65530] is too low
```

Fix:

```
sudo sysctl -w vm.max_map_count=262144
```

Issue: generate-indexer-certs.yml missing

Cause:

Using wrong Git branch.

Fix:

```
git checkout v4.8.2
```

Issue: .opendistro_security errors

Cause:

Certificates not generated.

Fix:

```
docker compose -f generate-indexer-certs.yml run --rm generator
```

Then restart:

```
docker compose down -v  
docker compose up -d
```

Issue: Port 5601 not working

Cause:

Dashboard is mapped to port 443.

Correct URL:

```
https://SERVER-IP
```

NOT:

```
http://SERVER-IP:5601
```

Issue: Docker Compose version warning

Warning:

```
the attribute `version` is obsolete
```

Fix:

Remove this line from docker-compose.yml:

```
version: '3.7'
```

11. Verify Open Ports

Check port 443:

```
sudo ss -tulnp | grep 443
```

Check port 9200:

```
sudo ss -tulnp | grep 9200
```

Check Wazuh API port:

```
sudo ss -tulnp | grep 55000
```

12. Final Working Architecture

Service	Port	Purpose
Wazuh Dashboard	443	Web UI
Wazuh Indexer	9200	Search engine
Wazuh Manager	1514/1515	Agent communication
Wazuh API	55000	API access

13. Final Result

After successful setup:

- Dashboard accessible in browser
- Login working
- Wazuh Manager healthy
- Wazuh Indexer healthy
- Docker containers running successfully
- Ready to add agents